In words: the time derivative of the expectation value of any observable **L** is given by $i/\hbar$ times the expectation value of the commutator of the Hamiltonian with **L**. Applying this principle to the velocity v, we find that

$$v = \frac{i}{2m\hbar}\langle\, [\, \mathbf{P}^2, \mathbf{X}\,]\, \rangle. \qquad (9.9)$$

Now, all we have to do is compute the commutator of $\mathbf{P}^2$ and **X**. A couple of simple steps shows that

$$[\, \mathbf{P}^2, \mathbf{X}\,] = \mathbf{P}[\, \mathbf{P}, \mathbf{X}\,] + [\, \mathbf{P}, \mathbf{X}\,]\mathbf{P}. \qquad (9.10)$$

This relation can be confirmed by expanding each commutator and spotting some obvious cancellations.

> **Exercise 9.2:** Prove Eq. 9.10 by expanding each side and comparing the results.

The last step uses the standard commutation relation

$$[\, \mathbf{P}, \mathbf{X}\,] = -i\hbar.$$

Substituting this into Eq. 9.10 and plugging that result into Eq. 9.9, we find that

$$v = \frac{\langle \mathbf{P} \rangle}{m}$$

or, perhaps more familiarly,

$$\langle \mathbf{P} \rangle = mv. \qquad (9.11)$$